

EVIDENCE · CANVAS · PROSE

# math

Boardroom-quality math layouts for mathematicians, quants, ML researchers, physicists, statisticians, and economists. KaTeX-rendered equations with persona-appropriate surround.

## The closed-form estimator.

LATTICE · MATH

$$\hat{\beta} = (X^{\top} X)^{-1} X^{\top} y$$

WHERE

- $\hat{\beta}$  — OLS coefficient vector
- $X$  — design matrix,  $n \times p$
- $y$  — response vector, length  $n$
- $X^{\top} X$  — Gram matrix,  $p \times p$ , must be invertible

## The log-likelihood we maximise.

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$$\ell(\beta) = \sum_{i=1}^n \left[ y_i \log \sigma(x_i^\top \beta) + (1 - y_i) \log(1 - \sigma(x_i^\top \beta)) \right]$$

WHERE

- $\ell$  — log-likelihood, concave in  $\beta$
- $\sigma$  — logistic link,  $\sigma(z) = 1/(1 + e^{-z})$
- $y_i$  — observed label,  $\in \{0, 1\}$
- $x_i$  — feature vector for observation  $i$

# Derivative of $f$ from first principles.

$$f(x + h) = f(x) + f'(x) h + O(h^2)$$

*Taylor expansion,  $n = 2$*

$$f(x + h) - f(x) = f'(x) h + O(h^2)$$

*subtract  $f(x)$*

$$\frac{f(x + h) - f(x)}{h} = f'(x) + O(h)$$

*divide by  $h \neq 0$*

$$\lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h} = f'(x)$$

**take the limit**

# Intermediate Value Theorem.

**DEFINITION.** A function  $f : [a, b] \rightarrow \mathbb{R}$  is *continuous* on  $[a, b]$  if  $\lim_{x \rightarrow c} f(x) = f(c)$  for every  $c \in [a, b]$ .

**THEOREM.** Let  $f$  be continuous on  $[a, b]$  and let  $y$  lie strictly between  $f(a)$  and  $f(b)$ . Then there exists  $c \in (a, b)$  with  $f(c) = y$ .

**PROOF.** Set  $S = \{x \in [a, b] : f(x) < y\}$ .  $S$  is non-empty and bounded; let  $c = \sup S$ . Continuity at  $c$  forces  $f(c) = y$ .  $\square$

## Frequentist vs Bayesian point estimate.

### FREQUENTIST

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$$\hat{\theta}_{\text{MLE}} = \arg \max_{\theta} p(y \mid \theta)$$

Maximises the likelihood — no prior. Uncertainty quantified by the sampling distribution of  $\hat{\theta}$  across hypothetical repeats.

### BAYESIAN

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$$\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} p(\theta \mid y)$$

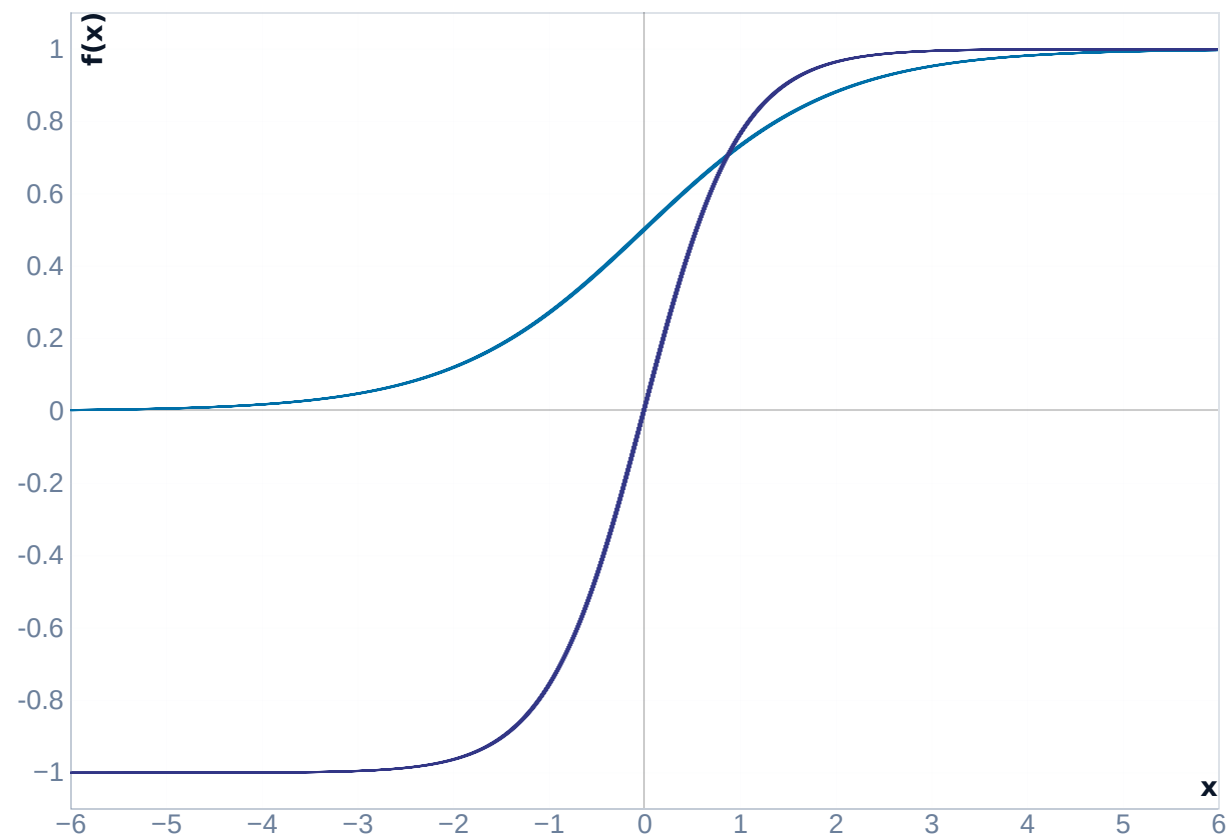
Maximises the posterior — conditions on the prior  $p(\theta)$ . Uncertainty is the posterior itself, no repeated sampling required.

# The sigmoid.

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$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

Maps  $\mathbb{R} \rightarrow (0, 1)$ . *S*-shaped,  $\sigma(0) = 0.5$ ,  
steepest slope at the origin.



# The design matrix $X$ .

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$$X = \begin{pmatrix} 1 & x_{11} & \cdots & x_{1p} \\ 1 & x_{21} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \cdots & x_{np} \end{pmatrix}$$

- `SHAPE` —  $n \times (p + 1)$
- `ROWS` — observations
- `COLS` — intercept +  $p$  features
- `RANK` — full-rank for OLS to have a unique solution
- `COLUMN 0` — all-ones, absorbs the intercept



## Effect of the treatment.

$$\hat{\beta} = 0.42 \pm 0.03$$

95% CI: [0.36, 0.48]

$p < 0.001$  ·  $n = 1,204$

*For every additional unit of exposure, the outcome rises by 0.42 SD — roughly an **8%** shift on the baseline. Effect size is the headline; the  $p$ -value just rules out chance.*

## LU decomposition.

$$\begin{pmatrix} 2 & 1 \\ 4 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$$

- $A$  — the original matrix being factorised
- $L$  — lower-triangular, unit diagonal
  - $U$  — upper-triangular
- `USE` — solve  $Ax = b$  by forward then back substitution

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## When NOT to reach for math.

Two

**display equations in the base layout**

The bare math layout is built around one hero equation. For side-by-side display, use

`math`  
`compare`

. For a derivation chain, use

`math`  
`derivation`

Stacking two

blocks in the base layout breaks the visual contract.

**Symbols without a legend.**

An equation with three undefined symbols is a puzzle, not a claim. Either every non-trivial symbol gets a legend entry, or the equation is simple enough that the audience knows it cold.

**ASCII math instead of KaTeX.**

Writing

`beta_hat`  
`=`  
`(X'X)^-1`  
`X'y`

as plain text bypasses the renderer. Always wrap math in

`$$...` (display)  
`$$` or

`$...`  
`$`

(inline) — KaTeX is the entire reason this layout exists.

#### RELATED COMPONENTS

## See also.

- `code` — the implementation, not the equation, is the argument
- `diagram` — the structure of the model, not its closed form
- `stats` — a row of statistical results without a single equation focus
- `content` — one inline equation inside a paragraph of prose